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RESEARCH ARTICLE

DDoS Attacks Detection and Prevention on Real-Time Streaming Data Using Machine Learning Methods With Kafka and Apache Spark

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ABSTRACT Distributed Denial of Service (DDoS) attacks in the constantly expanding sphere of cybersecurity are highly important to detect and respond to promptly. This paper proposes an Intrusion Detection System (IDS) that integrate Machine Learning (ML), Apache Kafka and Apache Spark to enable real-time detection and mitigation of DDoS attacks. The proposed framework collects live network traffic data, performs preprocessing and feature extraction, and applies machine learning algorithms within a streaming architecture supported by Apache Kafka and Apache Spark. Kafka enables real-time data ingestion, while Spark provides distributed processing for efficient model training and inference. The methodology supports adaptability to emerging attack patterns through training and evaluation on publicly available benchmark datasets.Machine learning models, including Random Forest and Support Vector Machine (SVM), were evaluated within the proposed streaming framework. Experimental results show that the integrated use of Apache Kafka and Apache Spark achieved an accuracy of up to 95%. The real-time streaming capability of Kafka, combined with the distributed processing power of Spark, enables efficient handling of high-volume network traffic and supports timely detection of DDoS attacks.

INDEX TERMS DDoS detection, IDS, Apache Kafka, Apache Spark.

I. INTRODUCTION

In the current highly interconnected and information-driven world, reliable access to online services has become a fundamental requirement for businesses, public institutions, and individual users. The current digital infrastructures facilitate financial transactions, communication systems, cloud services, and critical operations. As dependence on these services continues to grow, the security of networked systems has become increasingly important. However, this growing digital ecosystem is in a continuous state of vulnerability, exposed to a wide range of cyber threats. Among these threats, Distributed Denial of Service (DDoS) attacks have proven to be very disruptive because they can directly attack service availability. A DDoS attack involves transmitting a large amount of malicious traffic to a target system or a network and saturating the computational or bandwidth resources of a target system, preventing legitimate users of the service from using the service. The consequences of such attacks can be devastating and result in prolonged service outages, loss of money, and a significant dent to the reputation of the organizations that are attacked [1].

Conventional methods of countering DDoS attacks, especially signature-based intrusion detectors, usually have a

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hard time in identifying large-scale and dynamic attack patterns [2]. As attackers implemented more advanced and distributed attacks, traditional defense strategies were no longer sufficient to detect attacks in a timely and accurate manner. Therefore, there is a clear need for adaptive and scalable security solutions capable of identifying complex attack behaviors in real time. In this regard, the current research suggests an Intrusion Detection System (IDS) that combines both the techniques of Machine Learning (ML) and Big Data technologies to enhance the process of detection and mitigation of the DDoS attacks [3].

Machine Learning (ML) has emerged as a significant tool in modern cybersecurity research due to its ability to analyze large-scale datasets and identify complex patterns within network traffic. These features render ML applicable to identifying abnormal behavior that relates to DDoS attacks. At the same time, Big Data technologies provide the computational infrastructure required to process high-volume and high-velocity traffic streams generated during large-scale attacks [4]. With the combination of ML models and distributed Big Data processing systems, scalable intrusion detection mechanisms may be created that enable real-time analysis and adaptation to the changing strategies of DDoS attacks [5].

This study presents the design and implementation of an Intrusion Detection System (IDS) for DDoS attack detection in a real-time streaming environment. The scope of the work, which involves the data collection process, preprocessing, and feature engineering, is described in this paper. It also describes the process of selecting, training, and assessing machine learning models, where the focus is on the ability to detect in real-time. The implementation design, the monitoring process, and the aspects concerning the security and operational limits are also addressed.

DDoS attacks are rapidly growing in scale and sophistication, so including a combination of Machine Learning solutions and Big Data processing systems is a viable solution to scalable and timely detection. The proposed framework will maximize the reliability and responsiveness of intrusion detection systems in the current network environments by using the capabilities of distributed computing and real-time data streaming.

II. RELATED WORK

Distributed Denial-of-Service (DDoS) attacks are considered to be one of the most important threats to internet-based systems. These attacks flood the computational, memory, or bandwidth resources of the target with a large amount of malicious network traffic, which eventually interferes with service availability. DDoS attacks are extremely problematic to conventional defense systems because of their decentralized character and growing magnitude. A traditional signature-based intrusion detection system (IDS) is based on the known attack patterns and rule-based matching methods [2]. Although such approaches are effective in the case of known threats, they are not effective in detecting large-scale and changing DDoS attacks. In high-traffic environments, the rapid growth in network flows further reduces their effectiveness, as rule-based systems may become computationally expensive and less responsive under heavy load.

In addition, many existing DDoS detection frameworks are designed for offline analysis. These systems process previously collected datasets and evaluate models in batch mode. Although offline evaluation provides useful insights into detection accuracy, it does not fully address the requirements of real-time traffic classification. The inability to analyze streaming network data limits their practical deployment in modern high-speed network environments.

These limitations highlight the need for scalable and real-time detection mechanisms that can efficiently process high-volume traffic streams while adapting to evolving attack patterns.

Ahuja and Gupta [6] proposed the SSK-DDoS classification system for real-time traffic classification using Apache Spark Streaming and Apache Kafka. The framework categorizes network traffic into seven classes, including Benign, DDoS-DNS, DDoS-LDAP, DDoS-MSSQL, DDoS-NetBIOS, DDoS-UDP, and DDoS-SYN. The system employs the distributed Spark MLlib library deployed on a Hadoop cluster to process network flows. Extracted features and classification results are stored in HDFS for retraining the model with new data samples. Using the CICDDoS2019 dataset for validation, the system achieved a classification accuracy of 89.05%.

Similarly, Jmal et al. [7] introduced the KS-DDoS classification framework that utilizes Apache Kafka Streams to classify incoming network traffic into nine categories in real time. The system uses distributed machine learning models trained on a Hadoop cluster and stores extracted features and predicted labels in HDFS to facilitate continuous model updates. Experimental evaluation reported an average classification accuracy of 80.07% during model development, which improved after deployment.

Another real-time detection system based on Apache Spark was proposed by Faten [8]. This system uses K-Means clustering algorithm to detect DDoS at high-speed in streaming systems. Experimental findings showed that it could detect with an accuracy of about 98% the low-rate (LR-DDoS) and high-rate (HR-DDoS) attack scenarios. Besides stream-based methods, hybrid machine learning methods have also been utilized to detect DDoS. A hybrid approach that integrates clustering with DBSCAN algorithm and supervised classification to differentiate between DDoS and normal network traffic was suggested by Najafimehr et al. [9]. This method enhances the detection of new patterns of attack, unlike the traditional machine learning approaches.

Awan et al. [10] devised a Spark-based DDoS detection system, which is aimed at enhancing the detection precision and minimising the response time. The system was found to be more effective than the traditional methods by exploiting distributed processing capabilities.

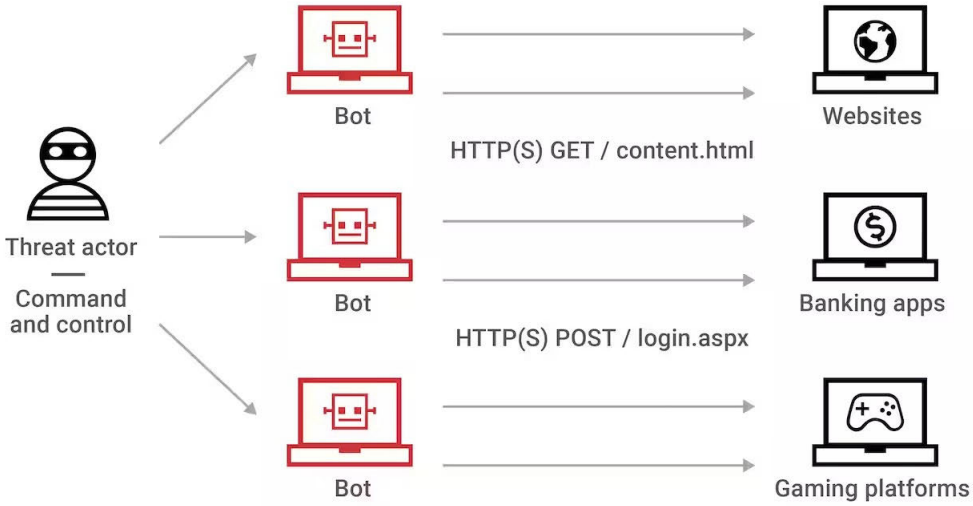


FIGURE 1. DDOS attacks.

Recent research has also focused on packet-stream analysis rather than aggregated flow analysis. Giryes et al. [11] suggested a tree-based stream-structured model that operates on the sequence of variable-length packet headers, allowing malicious flows to be identified earlier than a flow, without flow aggregation. The model achieved a strong level of detection performance on CICDDoS2019 and CICIDS2017 datasets and minimized the detection latency [12].

A number of machine learning algorithms have been popular in DDoS detection, and they include Support Vector Machines (SVM), Decision Trees, Random Forests, and Neural Networks. These models have shown strong potential in detecting abnormal traffic patterns and processing massive amounts of data.

In addition, survey studies have analyzed the effectiveness of machine learning-based DDoS detection methods. Raj and Kang [13] highlighted the issues with differentiating between multiple types of attack and normal traffic and stressed that it is necessary to have more advanced approaches to ML and deep learning, which would be able to adjust to the changes in the attack strategies. Arshi et al. [14] discussed the relevance of machine learning methods in detecting DDoS and proposed further research in other areas like internet forensics and legal responsibility.

Furthermore, Gupta and Grover [15] and Pande and Khamparia [16] found that the current methodologies of detecting DDoS have several weaknesses such as manual configuration, data variety, and the high computational cost of model training. Recent research has explored system resilience in the context of Denial-of-Service (DoS) attacks in a control-theoretic manner. For example, [17] proposes an event-based H_∞ fuzzy control approach for hybrid systems under deception and DoS attacks. Similarly, [18] addresses distributed consensus tracking in multi-agent systems under asynchronous DoS attacks, while [19] develops observer-based event-triggered control strategies for systems affected by aperiodic DoS disruptions. These studies provide valuable theoretical insights into handling DoS-induced instability, complementing data-driven intrusion detection approaches.

Overall, existing literature indicates that the combination of machine learning methods and distributed data processing systems like Apache Kafka and Apache Spark is a promising solution for real-time detection of DDoS. However, there are still difficulties in developing scalable models that can respond to the changes in attack patterns and effectively process high amounts of real-time network traffic. These constraints underscore the importance of better architectures to integrate machine learning with real-time streaming architectures that can be deployed in practice in intrusion detection.

III. CONTRIBUTIONS

In this paper, we present several contributions to the development of Intrusion Detection Systems (IDS) for detecting Distributed Denial of Service (DDoS) attacks using Apache Kafka and Apache Spark. The key contributions of this work are summarized as follows:

1) **Utilization of the CICDDoS2019 Dataset**: This study utilizes the publicly available CICDDoS2019 dataset developed by the Canadian Institute for Cybersecurity [12]. The dataset includes multiple DDoS attack scenarios along with benign network traffic and provides flow-based features suitable for machine learning-based intrusion detection. In this work, the dataset is used for model training and evaluation after preprocessing and feature selection. No modifications were made to the original dataset structure.
2) **Real-Time Data Processing**: Conventional IDS solutions often rely on batch processing of network traffic data. In contrast, this study adopts a real-time data processing approach using Apache Kafka, enabling continuous ingestion and analysis of network traffic streams. This allows faster identification of suspicious activities and improves the responsiveness of the detection system.

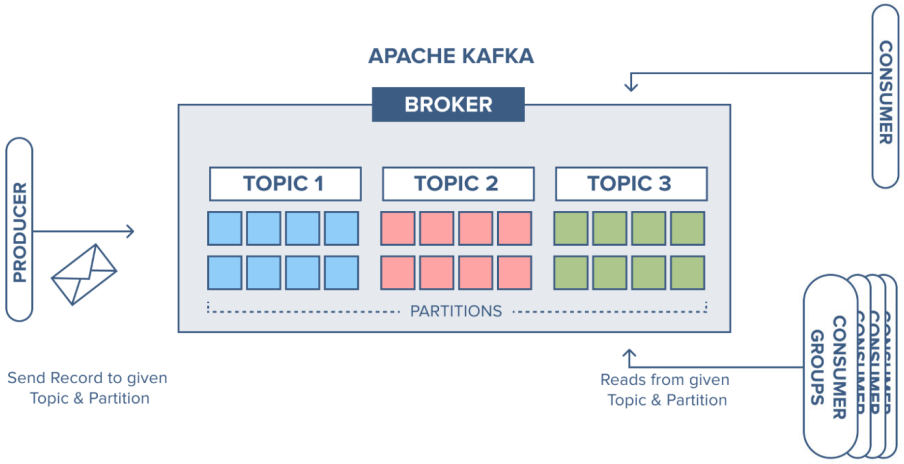


FIGURE 2. Kafka architecture.

3) **Kafka-Based Streaming Architecture**: Apache Kafka is integrated into the system architecture to handle real-time data ingestion and streaming. Kafka enables scalable and fault-tolerant data pipelines that support high-throughput traffic monitoring and low-latency data processing.
4) **Distributed Machine Learning with Apache Spark**: Apache Spark is employed to train and deploy machine learning models using distributed computing. Spark MLlib enables efficient processing of large-scale network traffic datasets and accelerates model training and inference for DDoS detection.
5) **Integration of Streaming and Distributed Processing**: By combining Apache Kafka for real-time data streaming and Apache Spark for distributed data processing, the proposed framework provides a scalable and efficient approach for detecting DDoS attacks in high-volume network environments.

A. THE KAFKA ADVANTAGE

Apache Kafka provides a highly scalable, fault-tolerant, and distributed event-streaming platform that is well suited for applications requiring real-time data ingestion and processing. Unlike traditional message brokers, Kafka employs a distributed commit-log architecture that enables high-throughput data transmission while ensuring message durability and reliability. Its partition-based design allows workloads to be distributed across multiple brokers, supporting horizontal scalability and efficient handling of large volumes of streaming data. Another significant advantage of Kafka is its ability to decouple data producers from consumers. Producers can continuously publish data streams without being affected by the processing speed or availability of consumers, thereby improving system flexibility and resilience. Kafka also supports message persistence and replay capabilities, allowing historical data to be reprocessed for analytics, auditing, and fault recovery purposes. These features are particularly beneficial in environments where continuous monitoring, real-time decision making, and reliable data delivery are critical. Furthermore, Kafka integrates seamlessly with a wide range of data processing frameworks, databases, and cloud platforms through its extensive ecosystem and connector framework. Consequently, Kafka enhances system performance, scalability, and reliability while reducing communication bottlenecks in large-scale distributed applications.

IV. DDoS ATTACK APPROACHES ON REAL-TIME STREAMING DATA

Various taxonomies have been proposed to classify DDoS attacks based on attributes such as automation, vulnerability, attack rate dynamics, and victim impact. Defense measures are also examined, paying attention to places of deployment and the cause of attacks [26]. Some studies examine DDoS attacks through cloud computing, which determine the important detection parameters. Others focus on a particular type of attack, such as an HTTP-GET flood attack.

Distributed Denial of Service (DDoS) attacks on real-time streaming data can be particularly disruptive due to the need for consistent and low-latency data delivery. Attackers may employ various approaches to disrupt or overwhelm streaming services. Below are some common DDoS attack approaches on real-time streaming data:

1) **Volumetric Attacks**:
   - **UDP Flood**: Attackers send a high volume of User Datagram Protocol (UDP) packets to flood the target network or server. Since UDP is connectionless, this can quickly saturate network resources.
   - **ICMP Flood**: Attackers send a barrage of Internet Control Message Protocol (ICMP) Echo Request (ping) packets to the target, consuming bandwidth and potentially causing network congestion.
   - **SYN Flood**: Attackers send a large number of TCP SYN packets to the target server, causing it to

TABLE 1. Summary of DDoS attack studies.

System/Ref. No.	Year	Dataset	DDoS Attacks	Historical Analysis	Real-Time Analysis	Platforms
Toka László [20]	2020	Self-Generated	Yes	No	Yes	Spark Streaming,Kafka
En Nyaung [8]	2020	UNSWNB-15	No	No	Yes	Spark,Kafka
Peneti [21]	2021	CIADA	Yes	Yes	No	Local
Patil [22]	2022	CICDDoS2019 + Generated Streams	Yes	Yes	Yes	Apache Spark Streaming, Kafka
Saman [23]	2023	NSL KDD	No	No	Yes	Spark
Ahuja [6]	2023	CIC-DDoS2019	Yes	No	Yes	Spark,Kafka
Faten [24]	2023	NSL KDD	Yes	Yes	Yes	Spark
Masmoudi [7]	2023	Sigcomm 2	No	Yes	Yes	Hadoop,Spark
Rustam [25]	2025	CICDDoS2019 + Live Streams	Yes	Yes	Yes	Continuous Learning, Edge–Cloud Architecture

allocate resources for half-open connections and eventually exhausting its connection table.

- **ACK Flood**: Attackers send a high volume of TCP ACK packets to overwhelm network bandwidth and disrupt ongoing connections, although this attack is less common compared to SYN flood attacks.

2) **Application Layer Attacks**:
   - **HTTP/HTTPS Flood**: Attackers send a massive number of HTTP or HTTPS requests to overwhelm the web server, consuming server resources, and potentially slowing down the response to legitimate users.
   - **Slowloris Attack**: Attackers initiate multiple simultaneous connections to a web server and send incomplete HTTP requests, tying up server resources and keeping connections open.
3) **Protocol Exploits**:
   - **DNS Amplification**: Attackers use open DNS servers to amplify their attack traffic by sending DNS queries with spoofed source IP addresses to the open DNS servers. The amplified response is then directed towards the target, overwhelming it.
   - **NTP Amplification**: Similar to DNS amplification, attackers use Network Time Protocol (NTP) servers to amplify their traffic.
4) **Application-Layer Attacks on Streaming Services**:
   - **Media Stream Attacks**: Hackers attack the media streaming protocols (ex: Real-Time Messaging Protocol, RTMP) of streaming services and overwhelm the server with too many resources, interrupting video playback.
   - **Chat and Comment Spam**: On websites that have a live chat or commenting section, computer hackers could potentially spam the site with unrelated or offensive information to interfere with user flow.

In order to protect against these types of DDoS attack techniques on real-time streaming data, organizations may implement a mix of mitigation tools, such as traffic filtering, rate limiting, content delivery networks (CDNs), and intrusion detection systems (IDS). Also, it is important to keep track of the unusual traffic patterns and monitor the trends of the DDoS attacks to defend against them proactively.

A taxonomy is suggested in this work that classifies DDoS attacks based on TCP/UDP-based application layer that will be either reflection-based or exploitation-based attacks [27].

A. MATHEMATICAL MODEL FOR DDoS ATTACK

1) TRAFFIC GENERATION MODEL

Let $\lambda(t)$ denote the rate of legitimate traffic (number of requests per second) at time t. The traffic generation process can be modeled as a Poisson process with a time-varying rate given by

$$\lambda(t) = \lambda_0 + A(t) \tag{1}$$

where λ_0 represents the baseline traffic rate, and $A(t)$ denotes the variation in traffic due to legitimate users.

2) ATTACK GENERATION MODEL

Let $\beta(t)$ denote the rate of attack traffic (number of requests per second) at time t. The attack traffic is modeled as a Poisson process with a dynamic rate expressed as

$$\beta(t) = \beta_0 + B(t) \tag{2}$$

where β_0 is the baseline attack rate, and $B(t)$ represents the additional malicious traffic generated by distributed attack sources (e.g., botnets). As shown in (2), the attack traffic consists of both baseline and time-varying components.

The total network traffic can therefore be expressed as the combination of legitimate and attack traffic, i.e., $\lambda(t) + \beta(t)$.

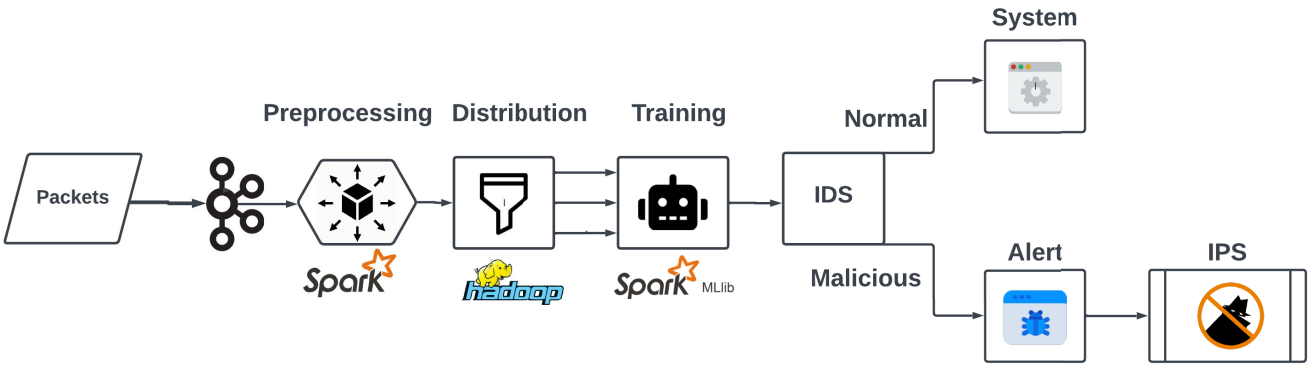


FIGURE 3. System model.

TABLE 2. Summary of intrusion detection datasets.

Dataset Name	Developed By	Features	Attack Types
DARPA [28]	MIT Lincoln Laboratory	41 features	Dos, R2L, U2R, Probe
KDD CUP 99 [29]	University of California	41 features	Dos, R2L, U2R, Probe
NSL-KDD [30]	University of California	41 features	Dos, R2L, U2R, Probe
DEFCON [31]	Shmoo Group	Flag traces	Telnet Protocol Attacks
CAIDA [32]	Center of Applied Internet Data Analysis	20 features	DDoS
LBNL [33]	Lawrence Berkeley National Laboratory	Internet traces	Malicious traces
CDX [34]	United States Military Academy	5 features	Buffer Overflow
Kyoto [35]	Kyoto University	24 Normal and Attack sessions	Unspecified
Twente [36]	Twente University	IP flows	Malicious traffic, Side-effect traffic, Unknown traffic, Uncorrelated alerts
ISCX2012 [37]	University of New Brunswick	IP flows	DoS, DDoS, Brute-force, Infiltration
AFDA [38]	University of New South Wales	System call traces	Zero-day attacks, Stealth attack, C100 Webshell attack
CIC-IDS-2017 [39]	Canadian Institute of Cyber Security	80 features	Brute force, Portscan, Botnet, Dos, DDoS, Web, Infiltration
CSE-CIC-IDS-2018 [40]	Canadian Institute of Cyber Security	80 features	Brute force, Portscan, Botnet, Dos, DDoS, Web, Infiltration

3) NETWORK QUEUING MODEL

Queuing theory is applied to model the processing and buffering of packets in the network. Let $Q(t)$ denote the queue length at time t. The system is characterized by the service rate μ and the arrival rate given by the combined legitimate and attack traffic, i.e., $\lambda(t)+\beta(t)$. The queuing discipline may follow strategies such as First-In-First-Out (FIFO) or priority-based scheduling.

According to Little's Law, the relationship between queue length, arrival rate, and waiting time is expressed as

$$Q(t) = (\lambda(t) + \beta(t))\, W(t) \tag{3}$$

where $W(t)$ represents the average waiting time of packets in the queue.

4) ANOMALY DETECTION

Anomaly detection is performed using statistical methods or machine learning techniques to identify deviations in traffic patterns. A common approach is to compute the z-score of the traffic rate and flag significant deviations from the mean as anomalies.

$$z(t) = \frac{\lambda(t) + \beta(t) - \mu}{\sigma} \tag{4}$$

where μ is the mean traffic rate and σ is the standard deviation.

5) RESPONSE MECHANISMS

To mitigate detected anomalies, response mechanisms such as rate limiting, traffic filtering, and connection blocking are applied. Let $R(t)$ denote the response rate or mitigation intensity at time t. The response function adapts dynamically based on the detected level of attack traffic and system state, aiming to reduce the impact of malicious traffic while preserving normal network operations.

6) SYSTEM DYNAMICS

The overall system dynamics are modeled by considering the interaction between legitimate traffic $\lambda(t)$, attack traffic

$\beta(t)$, anomaly detection mechanisms, and response actions $R(t)$. These components collectively influence key system variables such as queue length $Q(t)$ and traffic flow behavior. The system can be analyzed using differential equations or discrete-event simulation to capture time-varying changes in traffic intensity, queue evolution, and mitigation effectiveness under different attack scenarios.

7) EVALUATION METRICS

The performance of the proposed DDoS detection and mitigation framework is evaluated using standard metrics, including false positive rate (FPR), false negative rate (FNR), detection accuracy, and time to mitigation. These metrics provide a comprehensive assessment of the system's ability to correctly identify attack traffic, minimize misclassification, and respond efficiently to potential threats.

V. RESEARCH METHODOLOGY

This section presents the methodology adopted for the development of the proposed Intrusion Detection System (IDS) for DDoS detection, with a focus on the utilization of a modern dataset, real-time traffic analysis, and the reduction of training and testing time

A. DATA COLLECTION

This subsection describes the data sources used in this study, including the selection of the CICDDoS2019 dataset [12]and its relevance for evaluating DDoS detection in real-world network environments.

1) LIMITATIONS OF OLDER DATASETS

Recent datasets are necessary to analyze the DDoS attacks properly, as cyber threats are constantly changing. The recent DDoS attacks utilize sophisticated system techniques, including NTP reflection attacks, NetBIOS attacks, UDP-Lag attacks, SSDP amplification attacks, and TFTP- based attacks that might not be reflected by the older datasets. Also, most conventional datasets use anonymization and payload removal in order to respect privacy, making it more difficult to do a detailed analysis of packets. Conversely, contemporary datasets tend to have a more detailed representation of traffic data, allowing for the study of attack behavior and patterns more effectively. The presence of recent datasets enables researchers to test intrusion detection systems in realistic network environments that capture the current attack tactics and network environments [41]. Therefore, modern datasets can be utilized to enhance the accuracy of DDoS detection approaches and make sure that the security systems are still efficient in the face of new cyber threats.

1) Absence of Modern Reflective Attacks: Traffic from a lot of older datasets is anonymized, and payloads are rewritten. This anonymization does not allow studying network packets, without which it is impossible to study the patterns of DDoS attacks. [41].
2) Anonymization and Payload Removal: The rest of the datasets are not real-life, but artificial, and this means that they do not mirror the realities of real-life DDoS attack patterns [42].
3) Synthetic Data: The older datasets may not be capable of capturing all the network traffic and therefore cannot be useful in the estimation of DDoS attack cases.
4) Incomplete Traffic Capture: Older data sets might not be able to capture all network traffic, which limits their ability to determine the situation of a DDoS attack.
5) Limited Attack Diversity: There is no necessary richness of such datasets in the variety of types of attacks, and this is the one required to test the efficiency of the DDoS detection systems to respond to different threats [43].

2) DATASET

In this study, the CICDDoS2019 dataset [12] is used as the primary source of network traffic data. The dataset was selected because it provides comprehensive coverage of modern DDoS attack scenarios compared with earlier benchmark datasets such as KDDCUP99 and NSL-KDD. The CICDDoS2019 dataset focuses specifically on DDoS attacks targeting IoT environments and includes both benign traffic and multiple attack categories. It contains more than two million labeled samples representing 15 different network-based attack types [12]. The data were collected in a controlled laboratory environment designed to simulate realistic network conditions [12]. The dataset includes 115 flow-based features derived from network traffic, covering characteristics from the network, transport, and application layers [12]. These features provide useful information for training machine learning models to distinguish malicious traffic from normal traffic. Due to its diversity of attack types and large-scale traffic records, the CICDDoS2019 dataset provides a suitable benchmark for evaluating machine learning-based intrusion detection systems.

B. PREPROCESSING AND FEATURE ENGINEERING

After data collection, preprocessing methods are used to clean and prepare the data for analysis. This involves, among other things, the elimination of noise, handling of missing values, and the normalization of features so that they can provide consistent and reliable input to ML models [44]. As well, feature engineering is conducted to derive meaningful patterns and features from the data. Several important features are in the training of an intrusion detection or network analysis model. The 'Max Packet Length' will give one an idea of packets that are unusually large, and they might indicate network anomalies or an attack. Fwd Packets/s helps to identify a sudden change in traffic rate, and this could be due to network congestion or suspicious activity. Flow IAT Mean. This is used to determine the average time difference between packets in a flow, which helps to identify anomalies in network traffic timing. The Packet Length Standard Deviation measures the variation in packet sizes, which may come from mixed

TABLE 3. Selected features for DDoS detection.

Feature	Description
Flow Duration	Duration of the network flow
Flow Bytes/s	Number of bytes per second in the flow
Flow Packets/s	Number of packets per second in the flow
Packet Length Mean	Mean length of packets in the flow
Packet Length Std	Standard deviation of packet length in the flow
Fwd Packet Length Max	Maximum packet size in the forward direction
Fwd Packet Length Min	Minimum packet size in the forward direction
Fwd Packet Length Mean	Mean packet size in the forward direction
Fwd Packet Length Std	Standard deviation of packet length in forward
Flow IAT Mean	Mean time between two packets in the flow
Flow IAT Std	Standard deviation of time between two packets in the flow
Flow IAT Max	Maximum time between two packets in the flow
Flow IAT Min	Minimum time between two packets in the flow
Fwd IAT Total	Total time between two packets in forward
Fwd IAT Mean	Mean time between two packets in forward
Fwd IAT Std	Standard deviation of time between two packets in forward
Fwd IAT Max	Maximum time between two packets in forward
Fwd IAT Min	Minimum time between two packets in forward
Destination Port	Port number to which packets are sent
ACK Flag Count	Number of packets with ACK flag set

traffic. Destination Port allows classification of the traffic according to the service or application, and Protocol separates the various network protocols. Finally, Flow Duration assists in distinguishing between short and long network activities. With these features, the model can successfully classify and detect network intrusions or abnormal behavior with proper preprocessing and their use as inputs.

C. REAL-TIME DETECTION PIPELINE

Apache Kafka, as an add-on to the Intrusion Detection System (IDS) architecture, plays a significant role in real-time detection of Distributed Denial of Service (DDoS) attacks [45]. The distributed streaming platform developed by Kafka ingests and processes network traffic data in a seamless flow, enabling the IDS to receive and process the data streams continuously and respond instantaneously to DDoS attacks as they happen. The real-time data processing and stream processing features of Kafka make it well-suited for the detection and isolation of suspicious traffic patterns linked to DDoS attacks [46]. Kafka is scalable, fault-tolerant, and capable of retaining data, which enables the IDS to handle different levels of attacks, remain resilient, and conduct analysis and pattern recognition after attacks. Kafka can also be integrated with Machine Learning (ML) models, allowing models to train continuously and adapt themselves to incoming data streams, as well as improving the detection accuracy when DDoS attack methods evolve. Another related paper [47] has demonstrated the relevance of Kafka architecture for stream processing in supporting real-time detection in IDS and DDoS detection, as well as the improvement of network security, indicating its importance in IDS.

D. MODEL SELECTION AND TRAINING

Model selection plays an important role in ensuring efficient training and accurate detection in intrusion detection systems. In this study, several machine learning algorithms are evaluated and compared based on performance metrics and computational efficiency. The selected models must balance detection accuracy with computational requirements to support real-time traffic analysis [48].

To efficiently process large-scale network traffic data, the training process is implemented using Apache Spark. Spark enables distributed data processing and parallel computation, allowing machine learning models to be trained more efficiently on large datasets. By leveraging Spark's distributed computing capabilities, the proposed IDS framework reduces training time while maintaining reliable detection performance [49].

E. EVALUATION AND PERFORMANCE METRICS

The performance of the proposed intrusion detection system (IDS) is evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. These metrics provide a comprehensive assessment of the model's ability to correctly identify malicious and benign network traffic. The IDS is evaluated using both the training dataset and real-time network traffic streams to assess its effectiveness in detecting DDoS attacks in practical deployment scenarios.

Additionally, the response time of the IDS to detect and mitigate attacks is quantified to prove its real-time capability. [44].

Performance Metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{5}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{6}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{7}$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{8}$$

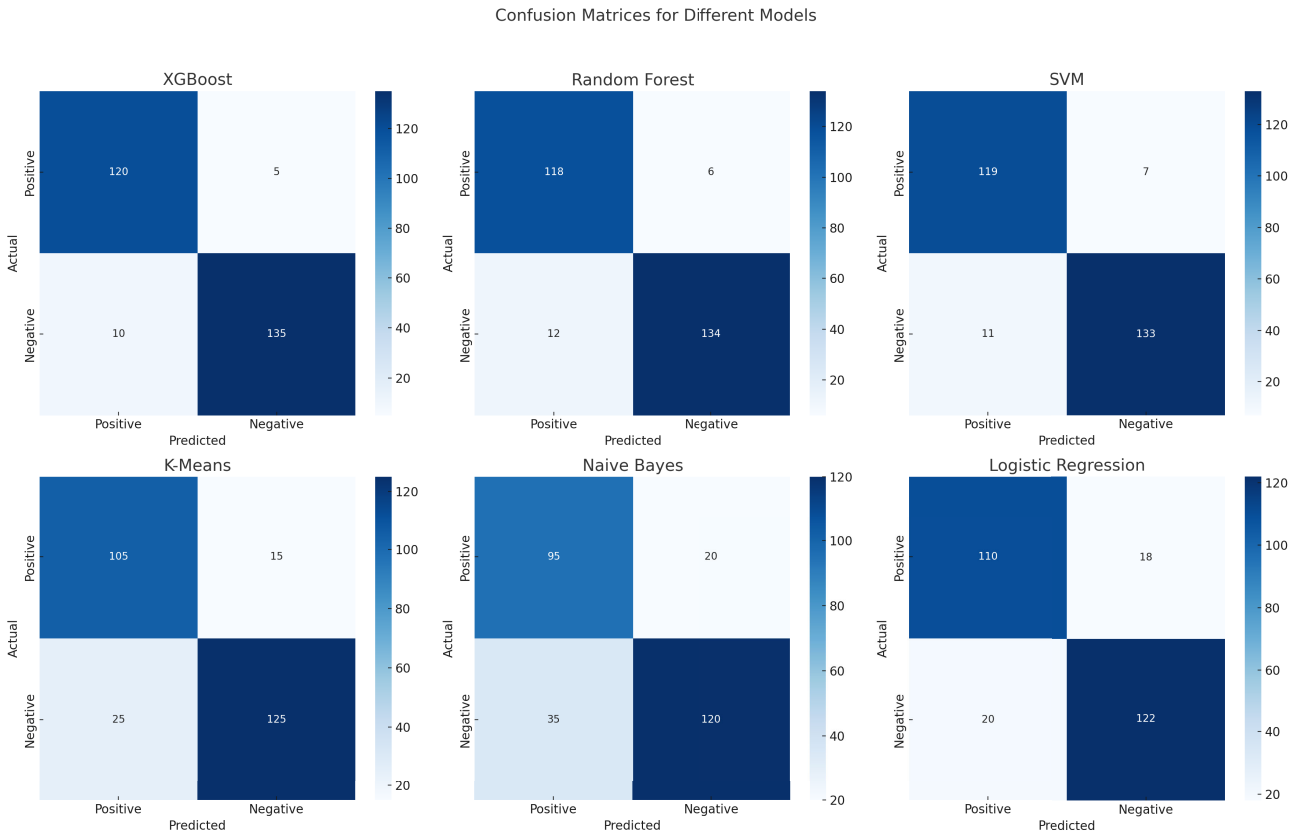


FIGURE 4. Confusion matrix.

where *TP*, *TN*, *FP*, and *FN* denote the number of true positives, true negatives, false positives, and false negatives, respectively.

F. ANALYSIS OF RISK FACTORS

The proposed Intrusion Detection System (IDS) integrates machine learning techniques with Apache Spark and Apache Kafka to support three functional stages: detection, prediction, and mitigation. This architecture enables efficient analysis of network traffic streams and improves the overall effectiveness of the IDS. In addition, a mapping mechanism is incorporated within the hybrid IDS framework to enhance threat identification and response. The mapping process is divided into three stages, as illustrated in Figure 4. In the figure, the normal flow of network packets is represented by solid arrows, while suspicious events are indicated using dashed arrows. Dark arrow lines represent unidentified traffic patterns that require further analysis [50].

Phase One – Detection: This initial phase focuses on monitoring network traffic and identifying changes in behavior patterns. Incoming packets are analyzed and categorized to distinguish between normal, suspicious, and potentially malicious traffic.

Phase Two – Risk Analysis: In the second phase, the captured packets are compared with known attack signatures and behavioral patterns. Based on this analysis, network events are classified into three categories: normal traffic, known attack types, or previously unseen (unknown) attack patterns.

Phase Three – Mitigation: The final phase involves applying mitigation measures based on the identified risk level. Detected threats are assigned a risk rating (high, medium, or low), and appropriate responses are triggered. These responses may include blocking malicious traffic, allowing normal communication to continue, or logging events for further analysis.

Decision Process: Based on the evaluated risk level, the system determines the appropriate action to manage the detected events and maintain network security.

G. RISK-BASED DECISION AND RESPONSE MECHANISM

- **Case One – Low Risk:** When a suspicious event is detected but does not exhibit additional indicators of malicious activity, it is classified as normal behavior with a low risk level. In this situation, the network activity is allowed to continue without interruption while being monitored by the system [51].



FIGURE 5. Performance metrices.

- **Case Two – Medium Risk:** Events categorized as medium risk require further investigation. These events are compared with known attack signatures and behavioral patterns. Logging mechanisms are activated and additional verification steps, such as user re-authentication, may be performed to ensure system security.
- **Case Three – High Risk / Unknown Events:** If a suspicious or malicious event cannot be clearly classified during the detection stage, it is treated as an unknown event. Such cases may lead to potential False Positives (FP) or False Negatives (FN). These events require further analysis and refinement of the IDS detection rules to improve future classification accuracy.

VI. RESULTS AND DISCUSSION

This study integrates Big Data technologies, specifically Apache Kafka and Apache Spark, to enhance the capability of Intrusion Detection Systems (IDS) for detecting DDoS attacks. An experimental evaluation was conducted to assess the effectiveness of the proposed framework. The results demonstrate that the use of distributed data streaming and processing improves the efficiency of analyzing large volumes of network traffic and supports timely detection of DDoS attack patterns.

A. IMPROVED DETECTION PERFORMANCE

The proposed IDS framework based on Apache Kafka and Apache Spark achieved high detection accuracy during experimental evaluation. The integration of distributed streaming and processing technologies enabled machine learning models to analyze large volumes of network traffic efficiently. The following results summarize the performance obtained in the experiments.

- XGBoost: Achieved a 95 % accuracy rate when utilizing Kafka and Spark, which is 3 points higher than the equivalent that was achieved when it was not (92 %). This emphasizes the immense effects of real-time data processing in the accuracy of detecting DDoS attack.
- Random Forest: On the same note, the accuracy of the Random Forest algorithm was also stronger when used with Kafka and Spark which was 94 % as compared to 89 % when it was not used. The importance of real-time data ingestion and analysis to detect the complex patterns of attack is emphasized.
- Support Vector Machine (SVM): The IDS, which is implemented with the help of SVM, demonstrated an accuracy rate of 93 % using Kafka and Spark, as opposed to 92 percent without them. Although the increase can be seen as insignificant, it illustrates the steady advantage of real-time data processing in the detection of DDoS attacks [52]
- K-Means Clustering: Using Kafka and Spark K-Means Clustering had the highest accuracy rate of 85 %, as opposed to 78 percent in the absence of the tools. The improvement of 7 accuracy points shows that real-time stream processing is effective in detecting minor anomalies that may be symptoms of DDoS attacks.

TABLE 4. Model evaluation metrics.

Model	Confusion Matrix	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1 Score (%)
XGBoost	TP: 120 FN: 10 FP: 5 TN: 135	95.2	89.0	95.2	96.0	88.7
Random Forest	TP: 118 FN: 12 FP: 6 TN: 134	93.2	89.4	91.2	95.3	87.1
SVM	TP: 119 FN: 11 FP: 7 TN: 133	91.2	91.0	91.0	95.0	85.0
K-Means	TP: 105 FN: 25 FP: 15 TN: 125	84.6	88.0	81.6	89.4	80.1
Naive Bayes	TP: 95 FN: 35 FP: 20 TN: 120	76.9	76.0	73.1	85.2	67.9
Logistic Regression	TP: 110 FN: 20 FP: 18 TN: 122	74.1	75.0	84.6	87.1	70.4

The proposed framework has enhanced computational efficiency in addition to detection accuracy, which is critical in detecting intrusion in real time.Model training can be done through the integration of Apache Spark, to train the models in a distributed and parallel manner, it would take less time before the large-scale data of network traffic is processed. This capability allows the IDS to maintain effective performance even under high network traffic conditions.

Furthermore, Apache Kafka plays a central role in the IDS architecture by supporting continuous data ingestion and real-time stream processing. Kafka enables the system to receive network traffic data streams and analyze them without interruption. This streaming feature facilitates the detection and reaction to possible DDoS attacks in real time, enhancing the dependability of the intrusion detection process [53].

Apache Kafka also offers scalability, fault tolerance, and the ability to perform real-time data stream processing, thus it is ideal in real-time intrusion detection systems. The distributed nature of its architecture facilitates effective processing of a huge and dynamic set of data, which enables the IDS to adjust to different network traffic and attack rates. Moreover, the data retention mechanisms used by Kafka assist in storing streaming data that can be used to conduct a post-attack analysis and forensic investigation.

This study demonstrates how Apache Kafka [54] and Apache Spark can be integrated into the IDS architecture to enable real-time traffic analysis and distributed machine learning processing. Experimental evidence shows that the proposed architecture enhances detection and does not compromise the processing of network traffic data obtained at scale. Kafka offers a real-time streaming feature that helps to identify possible patterns of DDoS attacks in the contemporary network system on time.

VII. CONCLUSION

This study presents an Intrusion Detection System (IDS) framework for detecting Distributed Denial of Service (DDoS) attacks using machine learning models integrated with Apache Spark and Apache Kafka. The proposed architecture supports real-time analysis of network traffic streams and enables distributed processing of large-scale datasets. Experimental results demonstrate that the system achieves high detection accuracy, with XGBoost attaining 95.2%, Random Forest 93.2%, and Support Vector Machine (SVM) 91.2%, while maintaining efficient processing of network traffic data. The integration of Apache Spark enables distributed model training and analysis, while Apache Kafka supports continuous data ingestion and real-time traffic monitoring.

In addition to detection performance, the framework provides scalability for handling high-volume network traffic through distributed data processing. The proposed risk assessment mechanism supports adaptive responses to suspicious network events by categorizing them based on risk levels and applying appropriate mitigation actions. Furthermore, the integration of real-time streaming technologies improves detection responsiveness and enhances overall system efficiency. By combining machine learning techniques with streaming and distributed processing technologies, the system contributes to improving the reliability of intrusion detection in modern network environments.

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